

The Transformer Architecture

Origins

The Transformer architecture was introduced in 2017 in the paper 'Attention Is All You Need' by Vaswani and colleagues at Google. It replaced the recurrent and convolutional layers that dominated earlier sequence models with a mechanism built entirely on attention. This change allowed for greater parallelism during training and became the foundation for modern large language models.

Self-Attention

The central idea of the Transformer is self-attention. For every token in a sequence, the model computes a query, a key, and a value vector. The attention score between two tokens is the dot product of one token's query with another token's key. These scores are passed through a softmax and used to take a weighted sum of the value vectors, letting each token gather information from every other token regardless of distance.

Multi-Head Attention

Rather than computing attention once, the Transformer runs several attention operations in parallel, called heads. Each head can focus on a different kind of relationship, for example syntax in one head and coreference in another. The outputs of all heads are concatenated and projected back to the model dimension.

Positional Encoding

Because self-attention has no inherent sense of order, the Transformer adds positional encodings to the input embeddings. These encodings inject information about the position of each token in the sequence, so the model can distinguish 'dog bites man' from 'man bites dog'.

Encoders, Decoders, and LLMs

The original Transformer had an encoder stack and a decoder stack. Many modern large language models, including the Llama family used by this project, are decoder-only models trained to predict the next token. Scaling these decoder-only Transformers to billions of parameters and training them on large text corpora produced the general-purpose language models that power tools like retrieval-augmented assistants.